Lecture 19: Autoregressive Linear Models

IEOR E4709: Statistic Analysis and Time Series
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1 Announcement

No in-person class on Wednesday, April 9, 2024. The video-recorded lecture will be broadcast and made available from the video library on Courseworks at 10:10am (usual class time).

2 Linear Time Series Analysis

Most of the material from this part of the class on Time Series is taken from the textbook "Analysis of financial Time Series" by Ruey Tsay.

The objective of financial time series analysis is to set up a family of probability models to represent data over time. Our goal is to estimate parameters of the model, and check the model for goodness of fit. Treating an asset return (e.g. the return r_t of a stock) as a collection of random variables over time, we have a time series $\{r_t\}_{t\geq 0}$. We will focus on linear time series analysis, where the data point at time t depends linearly or previous data points of interest. Linear time series analysis provides a framework for studying the dynamic structure of such a series.

The theories of linear time series usually discussed include stationarity, dynamic dependence, autocorrelation function, modeling and forecasting. Correlation plays an important role in understanding these models. In particular, correlations between the variable of interest and its past values are the focus of linear time series analysis. These correlations are referred to as *serial correlations* or *autocorrelations*. For instance, to test whether stock returns in different periods are uncorrelated, we want to verify that the auto-correlation is equal to zero. Next, we discuss how this can be achieved, but first we introduce some preliminary concepts and definitions.

A time series is *strictly stationary* if the joint distribution of $(r_{t_1}, r_{t_2}, \ldots, r_{t_k})$ is the same as the distribution of any other portion of the series $(r_{t_1+\tau}, r_{t_2+\tau}, \ldots, r_{t_k+\tau})$, where τ can be any integer. This is a very difficult condition to verify empirically. We will work with a weak version of stationarity, and assume that the return series of a stock $\{r_t\}$ is *weakly stationary*, i.e., $\mathbb{E}[r_t]$ and $cov(r_t, r_{t-l})$ exist and are time invariant, i.e., independent of t for any l.

We will use $\mu := \mathbb{E}[r_t]$ to denote the *common mean*, and $\gamma_l = cov(r_t, r_{t-l})$ to denote the lag-l auto-covariance of $\{r_t\}$.

Remark 2.1. *In finance, it is common to assume that an asset return series is weakly stationary. This assumption can be verified empirically if we have a sufficient number of historical returns available. For instance, we can divide the data into subsamples and then see if the mean and lag-l autocovariances are the same across subsamples.*

Consider the volatility of a stock. It is reasonable to expect (and empirically observed in data) that the volatility tomorrow will be correlated with the volatility today. However, will the volatility five months from now depend in any way on today's volatility? In order to answer questions of this sort, we need to define the autocorrelation function. The auto-covariance function is defined as $\mathbb{E}\left[(r_{t_1} - \mu_{t_1})(r_{t_2} - \mu_{t_2})\right]$. The autocorrelation function can be computed from the autocovariance function dividing by the standard deviations of r_{t_1} and r_{t_2} which corresponds to our usual definition of correlation. In other words, the autocorrelation function is given by $\mathbb{E}\left[(r_{t_1} - \mu_{t_1})(r_{t_2} - \mu_{t_2})\right]/(\sigma_{t_1}\sigma_{t_2})$. In the case of a stationary time series, the *lag l auto-correlation* $\rho_l = corr(r_t, r_{t+l})$ is given by $\frac{\gamma_l}{\gamma_0}$, when $\gamma_0 \neq 0$.

Clearly, we have that $\gamma_l = \gamma_{-l}$, and γ_0 is the common variance of $\{r_t\}_{t\geq 0}$. Hence, let us use $\Gamma_n := [\gamma_{i-j}]_{i,j=1}^n$ to denote the covariance matrix of $r_{t+1}, \ldots, r_{t+n}$. The matrix Γ_n is positive semidefinite (its smallest eigenvalue is nonnegative).

In practice, the auto-covariance and auto-correlation functions are unknown and have to be estimated. Consider observations $\{r_1, r_2, \ldots, r_T\}$ of a weakly stationary time series. The *sample auto-covariance* of the observations is

$$\hat{\gamma}_l = \frac{1}{T}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}), \qquad 0 \leq l < T, \tag{1}$$

where $\bar{r} := \frac{1}{T}\sum_{t=1}^{T} r_t$ is the sample mean. The *sample covariance* matrix of $r_{t+1}, \ldots, r_{t+n}$ is $\hat{\Gamma}_n := [\hat{\gamma}_{i-j}]_{i,j=1}^n$. It holds that if $\hat{\gamma}_0 > 0$, then $\hat{\Gamma}_n$ is invertible. The sample auto-correlation is given by $\hat{\rho}_l = \frac{\hat{\gamma}_l}{\hat{\gamma}_0}$. Alternatively, one could use

$$\hat{\gamma}_l = \frac{1}{T-l}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}), \qquad 0 \leq l < T, \tag{2}$$

as an estimate of the autocovariance function. Both estimator in (1) and (2) are consistent. The estimator (1) produces positive definite autocovariance matrices in any sample. By contrast, the estimator (2) does not (necessarily) produce positive definite sequences. One can use either, and there is little difference when T is large. I will use the estimator (1) in this class, which is typically the most commonly used.

Remark 2.2. *It is worth observing that both estimators* (1) *and* (2) *are biased. If we divide by* $T - l - 1$ *instead* $T - l$*, i.e.,*

$$\hat{\gamma}_l = \frac{1}{T-l-1}\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r}),$$

one can then show that $\hat{\gamma}_l$ *is an unbiased estimator of the lag* l *autocovariance* γ_l*.*

Example 1. *Consider a time series of length* 2*, where* $x_t = 0$ *and* $x_{t+1} = 1$*. It is then simple to see that the matrix* $\hat{\Gamma}$*, whose entries are computed dividing by* $T - l = 2 - 1 = 1$ *in this case, is given by* $\hat{\gamma}(1,1) := \hat{\gamma}_{1-1} = 1/4$*,* $\hat{\gamma}(1,2) := \hat{\gamma}_{1-2} = -1/4$*,* $\hat{\gamma}(2,1) := \hat{\gamma}_{2-1} = -1/4$*,* $\hat{\gamma}(2,2) := \hat{\gamma}_{2-2} = 1/4$*, which is singular. To compute these entries, we have used the formula*

$$\frac{1}{T-h}\sum_{t=1}^{T-h}(x_{t+h} - \bar{x})(x_t - \bar{x})$$

Instead, the matrix $\hat{\Gamma}$, whose entries are computed dividing by $T = 2$ is given by $\hat{\gamma}(1,1) = 1/4$, $\hat{\gamma}(1,2) = -1/8$, $\hat{\gamma}(2,1) = -1/8$, $\hat{\gamma}(2,2) = 1/4$, which has eigenvalues 3/8 and 1/8, and thus it is positive-definite. To compute these entries, we have used the formula

$$\frac{1}{T}\sum_{t=1}^{T-h}(x_{t+h} - \bar{x})(x_t - \bar{x})$$

3 Randomness of a Time Series

With many time series, a trend or seasonal pattern may be evident from the time plot indicating that the series is not random. The lack of randomness may be evident by looking at the short term autocorrelation. For instance, if the time series alternates up and down around the mean in successive values, the lag-1 autocorrelation will be negative. On the other hand, if an observation above the mean tends to be followed by several other observations that are above the mean, then the autocorrelations for lags k will be positive (or, vice-versa, if an observation below the mean tends to be followed by successive observations that are also below the mean is an indication of positive lag k autocorrelations). If a time series exhibits these types of correlations, then the series is not completely random. One of the goals of time series analysis is to model this sort of autocorrelation. However, before pursuing this more complicated modeling, it would be useful to know if the series is completely random or not.

Consider the lag l-autocorrelation, which we recall is defined as

$$\hat{\rho}_l = \frac{\sum_{t=l+1}^{T}(r_t - \bar{r})(r_{t-l} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}.$$

If $\{r_t\}$ is an i.i.d. sequence such that $\mathbb{E}\left[r_t^2\right] < \infty$, then it can be shown that $\hat{\rho}_l$ is asymptotically normal with mean zero and variance $1/T$ for any fixed positive integer l; see the book by Brockwell and Davis 1991, Theorem 7.2.2, for a proof of this result. This result can be used to test, for instance, the null hypothesis H_0 that $\rho_1 = 0$ versus the alternative hypothesis H_a that $\rho_1 \neq 0$. Under the null hypothesis, $\sqrt{T}\hat{\rho}_1$ follows asymptotically the normal distribution.[1]

Ljung and Box (1978) introduced the statistics

$$Q(m) = T(T+2)\sum_{l=1}^{m}\frac{\hat{\rho}_l^2}{T-l},$$

where T is the sample size, as a test statistic for the null hypothesis $H_0 : \rho_1 = \rho_2 = \cdots = \rho_m = 0$ against the alternative hypothesis that $H_a : \rho_i \neq 0$ for some $i = 1, \ldots, m$. If $\{r_t\}_{t\geq 0}$ is an i.i.d. sequence satisfying certain moment conditions, we have that the asymptotic distribution of $Q(m)$ under the null hypothesis H_0 is a χ^2 distribution with m degrees of freedom. Hence, the test rejects if $Q(m) > \chi^2_{m,1-\alpha}$, where $\chi^2_{m,1-\alpha}$ is the $1-\alpha$ quantile of a χ^2 distribution with m degrees of freedom.[2]

[1] More generally, if the time series $\{r_t\}$ is weakly stationary satisfying $r_t = \mu + \sum_{i=0}^{p}\phi_i a_{t-i}$, where $\phi_0 = 1$ and $\{a_j\}$ is a sequence of i.i.d. zero-mean random variables, then $\hat{\rho}_l$ is asymptotically normal with mean zero and variance $(1 + 2\sum_{i=1}^{p}\rho_i^2)/T$ for $l > p$, where ρ_i is the actual lag i autocorrelation coefficient.

[2] See the paper by Ljung and Box (1978): "On a Measure of a Lack of Fit in Time Series Models", Biometrika, Vol. 65, No. 2, pp. 297-303 for the details.

4 AR(1) Model

Autoregressive (AR) models provide a representation for random time-varying processes arising in finance and economics. The autoregressive model specifies that the response (output) variable depends linearly on its own previous values and on a stochastic term (an unpredictable term). The model takes the form of a stochastic difference equation.

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t, \tag{3}$$

where $\{a_t\}$ is a white noise sequence, i.e., an i.i.d. sequence with zero mean and variance σ_a^2. The terms ϕ_0 and ϕ_1 are constants. Next, let us identify the conditions for the weak stationarity of the time series. Assuming that the time series $\{r_t\}$ is weakly stationary, we have that $\mathbb{E}[r_t] = \mu$ and $cov(r_t, r_{t-l}) = \gamma_l$. Let $\mathcal{F}_t$ be the sigma algebra generated by the return observations $r_1, r_2, \ldots, r_t$. We can compute the conditional mean given by

$$\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right] = \phi_0 + \phi_1 r_{t-1}, \qquad Var\left[r_t \middle| \mathcal{F}_{t-1}\right] = \sigma_a^2$$

The unconditional mean is given by

$$\mathbb{E}[r_t] = \mathbb{E}\left[\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right] = \phi_0 + \phi_1 \mathbb{E}[r_{t-1}]. \tag{4}$$

As $\{r_t\}$ is weakly stationary, we have that $\mathbb{E}[r_t] = \mu$ for any t. Using Eq. (4), we deduce $\mu = \phi_0 + \phi_1 \mu$, or equivalently

$$\mu = \frac{\phi_0}{1 - \phi_1}. \tag{5}$$

We can also compute the unconditional variance, given by

$$Var(r_t) = \mathbb{E}\left[Var\left(r_t \middle| \mathcal{F}_{t-1}\right)\right] + Var\left(\mathbb{E}\left[r_t \middle| \mathcal{F}_{t-1}\right]\right) = \sigma_a^2 + \phi_1^2 Var(r_{t-1}). \tag{6}$$

Because r_t is weakly stationary, i.e., $Var(r_t) = \gamma_0$, we can use Eq. (6) to deduce $\gamma_0 = \sigma_a^2 + \phi_1^2 \gamma_0$, hence

$$\gamma_0 = \frac{\sigma_a^2}{1 - \phi_1^2}, \tag{7}$$

provided that $\phi_1^2 < 1$. The requirement that $\phi_1^2 < 1$ results from the fact that the variance of a random variable is bounded and nonnegative. Consequently, the weak stationarity of an $AR(1)$ model implies that $-1 < \phi_1 < 1$, i.e., $|\phi_1| < 1$. It can be shown that this condition is also sufficient for weak stationarity, hence the *necessary and sufficient condition for the AR(1) model* (3) *to be weakly stationary is* $|\phi_1| < 1$. *See Section 2.4.1 of Ruey Tsay's textbook, Box and Jenkins, 1976, Ch. 3 or Pagano, 1973, for the formal proof.*

Using Eq. (5), we can rewrite the AR(1) model as

$$r_t - \mu = \phi_1 (r_{t-1} - \mu) + a_t,$$

or, equivalently, as

$$r_t = (1 - \phi_1)\mu + \phi_1 r_{t-1} + a_t.$$

This model is often used in finance, and ϕ_1 measures the persistence of an AR(1) time series, where high persistence means that past shocks have a strong impact on the current value of the variable.

We can also compute the auto-correlation function of the AR(1) time series. Using that r_t is a weakly stationary time series, we have

$$\gamma_l = cov(r_t, r_{t-l}) = \phi_1 cov(r_{t-1}, r_{t-l}) + cov(a_t, r_{t-l}) = \phi_1 \gamma_{l-1},$$

which admits the solution

$$\gamma_l = \phi_1^l \gamma_0, \qquad l \geq 1.$$

Because $\rho_0 = 1$, the lag-l auto-correlation function (ACF) is then given by

$$\rho_l = \phi_1^l, \tag{8}$$

where $l \geq 1$. The ACF of the AR(1) times series decays exponentially to zero, as the lag l goes to infinity.

4.1 Estimation of AR(1) Model

Suppose r_t, $t = 1, \ldots, T$, are observed. In order to estimate the parameters of the AR(1) model, we can proceed as follows:

- Estimate ϕ_1 by matching the population and the sample lag-1 autocorrelation, i.e. set $\hat{\phi}_1 := \hat{\rho}_1$, where $\hat{\rho}_1$ is the sample lag-1 autocorrelation.
- Using Eq. (5), i.e., $\mu = \frac{\phi_0}{1-\phi_1}$, we can estimate ϕ_0. This yields $\hat{\phi}_0 = \hat{\mu}(1 - \hat{\phi}_1)$, where $\hat{\mu}$ is the sample mean.
- Estimate σ_a^2 using Eq. (7), by matching the population and sample variance. This yields $\hat{\sigma}_a^2 = \hat{\gamma}_0(1 - \hat{\phi}_1^2)$, where $\hat{\gamma}_0$ is the sample variance.

An alternative estimation method is the maximum likelihood estimation, which can be used if we have knowledge of the returns distribution. Assume that, given r_j, $j = 0, \ldots, t-1$, then r_t is normally distributed with mean $\phi_0 + \phi_1 r_{t-1}$ and variance σ_a^2. We can then use maximum likelihood estimation to estimate the parameters. The likelihood of r_t, $t = 1, \ldots, T$, given r_0 is:

$$\log L(r_{1:T} | \phi_0, \phi_1, \sigma_a, r_0) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\sigma_a^2 - \frac{1}{2\sigma_a^2}\sum_{t=1}^{T}\left(r_t - (\phi_0 + \phi_1 r_{t-1})\right)^2.$$

We can then find the estimates by maximizing the conditional log-likelihood. It can be shown that the estimator is the same as the OLS estimator in the linear regression model where r_{t-1}, $t = 1, \ldots, T$ are the explanatory variables, and r_t, $t = 1, \ldots, T$ are the response variables. When T is large, the choice of r_0 does not have a big impact on the parameter estimates.

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